

90

FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

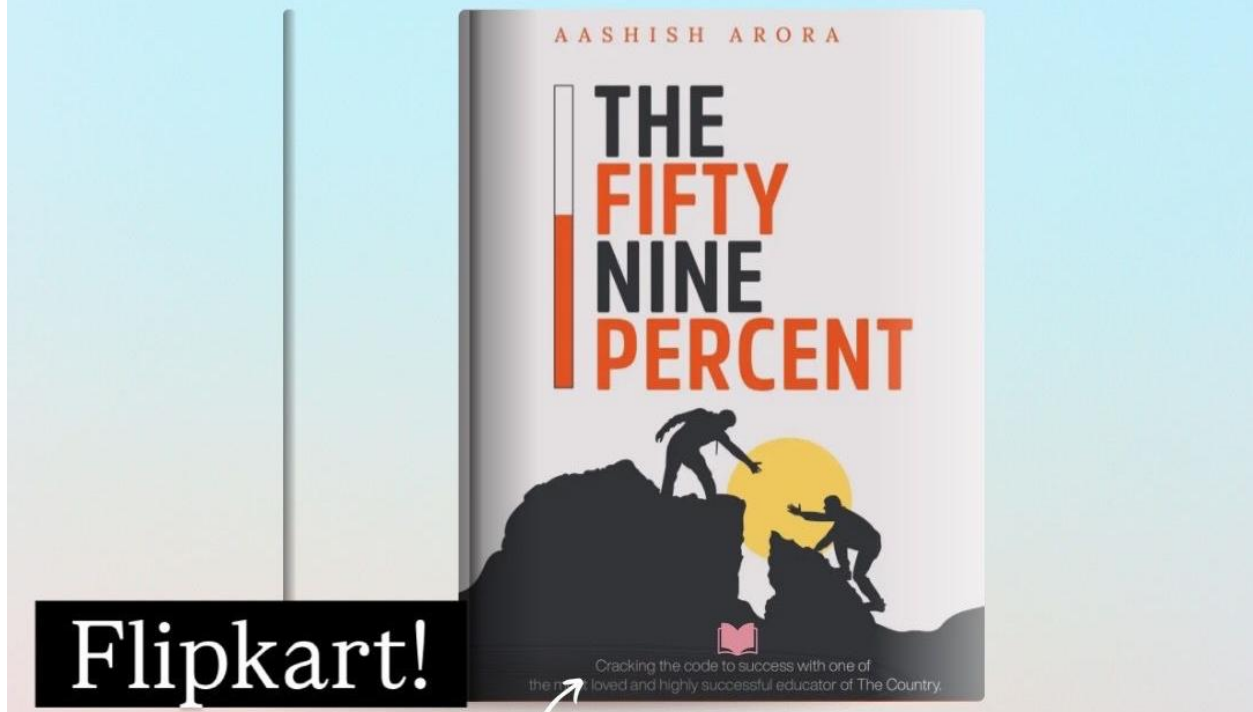


/STUDIFIEDPDF

Subscribe to
STUDIFIEDTM

**THE BOOK YOU
LOVED IS BACK**

Only at Rs.199/-



Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

yes
OFFICER

ALL COURSES UNDER ONE SUBSCRIPTION'

AASHISH ARORA EXCLUSIVE

MAINS COURSES

- MAINS FIRST
- THE MATHS HERO PRO
- MAINS SETU
- MAINS D.I SMARTBOOK
- LEVEL UP 1.0, 2.0
- MAINS MADE EASY 1.0, 2.0
- MAINS SARTHI
- OPERATION MAINS

FOUNDATION COURSES

- THE MATHS HERO 2026 (NEW)
- THE MATHS HERO 2025
- THE MATHS HERO 2024
- UDAAN D.I
- GANIT 1.0, 2.0
- LAKSHYA COURSE

USE CODE :- TT10

FOR MAXIMUM DISCOUNT



CONTENTS

1. SIMPLIFICATION & APPROXIMATION	9
2. ARITHMETIC WORD PROBLEMS	22
3. QUADRATIC EQUATIONS	43
4. WRONG NUMBER SERIES	59
5. MISSING NUMBER SERIES	70
6. DATA INTERPRETATION	82

1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $\sqrt{4096} \times 13 - 72.72\% \text{ of } 847 + \frac{5}{8} \text{ of } 928 = ? \times 8$

- A. 95.5
- B. 97.5
- C. 99.5
- D. 101.5
- E. None of these

2. $? + 37.5\% \text{ of } 1280 = \sqrt[3]{175616} + 24^2 - 18 \times 21$

- A. -216
- B. -226
- C. -236
- D. -246
- E. None of these

3. $\frac{3}{7} \text{ of } 1029 + 22.22\% \text{ of } 729 - \sqrt{1156} = ? + 41.66\% \text{ of } 624$

- A. 309
- B. 299
- C. 315
- D. 305
- E. None of these

4. $12.5\% \text{ of } (3840 + 1600) - \frac{5}{13} \text{ of } 1612 = 4 \times ? + 48$

- A. 5
- B. 2
- C. 4
- D. 3
- E. None of these

5. $\sqrt{(81 \times 64)} + 5^3 - 68.75\% \text{ of } 352 = ? \div 6$

- A. -260
- B. -270
- C. -280
- D. -250
- E. None of these

6. $2\frac{1}{4} \text{ of } 176 + 3\frac{3}{5} \text{ of } 125 - 56.25\% \text{ of } 640 = ? + 218$

- A. 268
- B. 258
- C. 272
- D. 264
- E. None of these

7. $(? \times 7) - 31.25\% \text{ of } 896 = 14^2 + 20 \times 21$

- A. 124

B. 130

C. 126

D. 128

E. None of these

8. 83.33% of 792 + 18.18% of 1089 + $\sqrt[3]{2744} = ? \times 8$

A. 107

B. 109

C. 105

D. 111

E. None of these

9. 20% of 45% of 1600 + $(17^2 - 11^2) \div 4 = ?$

A. 176

B. 192

C. 182

D. 186

E. None of these

10. $?^2 + 72.72\%$ of 484 = $28^2 - \sqrt[3]{32768}$

A. 20

B. 16

C. 22

D. 18

E. None of these

11. $1440 \div 12 \div 5 + 756 \div 9 \div 7 + 62.5\%$ of 384 = $? + 185$

A. 93

B. 91

C. 89

D. 87

E. None of these

12. $7\frac{1}{2}\%$ of 6400 + $3\frac{1}{8}\%$ of 9600 + 11.11% of 999 = $? \times 11$

A. 75

B. 78

C. 81

D. 84

E. None of these

13. $\sqrt{?} - 25\%$ of 128 = 58.33% of 432 - 16^2

A. 676

B. 841

C. 729

D. 784

E. None of these

14. $\sqrt{(64 \times 36)} + \sqrt[3]{(125 \times 216)} + 43.75\%$ of 512 = $? + 91$

A. 207

B. 215

C. 211

D. 209

E. None of these

15. $12 \times \sqrt{1369} - 16 \times \sqrt{625} + 23 \times \sqrt{196} = ? \times 6$

A. 61

B. 57

C. 63

D. 59

E. None of these

16. ?% of 960 + 125% of 480 = 225% of 640 - 360

A. 45

B. 52

C. 48

D. 50

E. None of these

17. $[(52^2 - 28^2) \div 16] + \frac{5}{6}$ of 504 - 32.5% of 400 = ? + 77

A. 331

B. 333

C. 329

D. 335

E. None of these

18. $9^2 + 8^3 - 6.25\%$ of 1024 = $(? \div 4) + 17^2$

A. 920

B. 980

C. 960

D. 940

E. None of these

19. 41.66% of 1728 - 33.33% of 1080 + 14.28% of 2450 = ? $\times$ 5

A. 142

B. 136

C. 146

D. 138

E. None of these

20. $(3\frac{1}{4} + 2\frac{3}{5}) \times 80 - 68.75\%$ of 640 = ?

A. 24

B. 32

C. 22

D. 28

E. None of these

Answers:

1. C

2. B

3. A

4. D

5. B

6. A

7. D

8. B

9. D

10. A

11. B

12. C

13. D

14. C

15. A

16. D

17. B

18. C

19. A

20. D

Solutions:

$$1. \sqrt{4096} \times 13 - 72.72\% \text{ of } 847 + \frac{5}{8} \text{ of } 928 = ? \times 8$$

$$72.72\% = \frac{8}{11}, \sqrt{4096} = 64, \text{ and } \frac{5}{8} \text{ of } 928 = 580$$

$$\text{So, } 64 \times 13 - \frac{8}{11} \times 847 + 580 = ? \times 8$$

$$832 - 616 + 580 = ? \times 8$$

$$796 = ? \times 8$$

$$? = 796 \div 8 = 99.5$$

$$2. ? + 37.5\% \text{ of } 1280 = \sqrt[3]{175616} + 24^2 - 18 \times 21$$

$$37.5\% = \frac{3}{8}, \sqrt[3]{175616} = 56, \text{ and } 24^2 = 576$$

$$\text{So, } ? + \frac{3}{8} \times 1280 = 56 + 576 - 378$$

$$? + 480 = 254$$

$$? = 254 - 480 = -226$$

$$3. \frac{3}{7} \text{ of } 1029 + 22.22\% \text{ of } 729 - \sqrt{1156} = ? + 41.66\% \text{ of } 624$$

$$\frac{3}{7} \text{ of } 1029 = 441, 22.22\% \text{ of } 729 = 162, \text{ and } \sqrt{1156} = 34$$

$$41.66\% = \frac{5}{12}, \text{ so } 41.66\% \text{ of } 624 = 260$$

$$441 + 162 - 34 = ? + 260$$

$$569 = ? + 260$$

$$? = 309$$

$$4. 12.5\% \text{ of } (3840 + 1600) - \frac{5}{13} \text{ of } 1612 = 4 \times ? + 48$$

$$12.5\% = \frac{1}{8} \text{ and } \frac{5}{13} \text{ of } 1612 = 620$$

$$\text{So, } \frac{1}{8} \times 5440 - 620 = 4 \times ? + 48$$

$$680 - 620 = 4 \times ? + 48$$

$$60 = 4 \times ? + 48$$

$$4 \times ? = 12$$

$$? = 3$$

$$5. \sqrt{(81 \times 64)} + 5^3 - 68.75\% \text{ of } 352 = ? \div 6$$

$$\sqrt{(81 \times 64)} = 72, 5^3 = 125, \text{ and } 68.75\% = \frac{11}{16}$$

$$\text{So, } 72 + 125 - \frac{11}{16} \times 352 = ? \div 6$$

$$72 + 125 - 242 = ? \div 6$$

$$-45 = ? \div 6$$

$$? = -270$$

$$6. 2\frac{1}{4} \text{ of } 176 + 3\frac{3}{5} \text{ of } 125 - 56.25\% \text{ of } 640 = ? + 218$$

$$2\frac{1}{4} \text{ of } 176 = 396, 3\frac{3}{5} \text{ of } 125 = 450, \text{ and } 56.25\% \text{ of } 640 = 360$$

$$\text{So, } 396 + 450 - 360 = ? + 218$$

$$486 = ? + 218$$

$$? = 268$$

$$7. (? \times 7) - 31.25\% \text{ of } 896 = 14^2 + 20 \times 21$$

$$31.25\% \text{ of } 896 = 280 \text{ and } 14^2 + 20 \times 21 = 196 + 420 = 616$$

$$\text{So, } ? \times 7 - 280 = 616$$

$$? \times 7 = 896$$

$$? = 128$$

$$8. 83.33\% \text{ of } 792 + 18.18\% \text{ of } 1089 + \sqrt[3]{2744} = ? \times 8$$

$$83.33\% \text{ of } 792 = 660, 18.18\% \text{ of } 1089 = 198, \text{ and } \sqrt[3]{2744} = 14$$

$$\text{So, } 660 + 198 + 14 = ? \times 8$$

$$872 = ? \times 8$$

$$? = 109$$

$$9. 20\% \text{ of } 45\% \text{ of } 1600 + (17^2 - 11^2) \div 4 = ?$$

$$45\% \text{ of } 1600 = 720 \text{ and } 20\% \text{ of } 720 = 144$$

$$17^2 - 11^2 = 289 - 121 = 168$$

$$\text{So, } 144 + 168 \div 4 = ?$$

$$144 + 42 = ?$$

$$? = 186$$

$$10. ?^2 + 72.72\% \text{ of } 484 = 28^2 - \sqrt[3]{32768}$$

$$72.72\% = \frac{8}{11} \text{ and } \sqrt[3]{32768} = 32$$

$$\text{So, } ?^2 + \frac{8}{11} \times 484 = 28^2 - 32$$

$$?^2 + 352 = 784 - 32$$

$$?^2 + 352 = 752$$

$$?^2 = 400$$

$$? = 20$$

$$11. 1440 \div 12 \div 5 + 756 \div 9 \div 7 + 62.5\% \text{ of } 384 = ? + 185$$

$$1440 \div 12 \div 5 = 24 \text{ and } 756 \div 9 \div 7 = 12$$

$$62.5\% \text{ of } 384 = 240$$

$$\text{So, } 24 + 12 + 240 = ? + 185$$

$$276 = ? + 185$$

$$? = 91$$

$$12. 7\frac{1}{2}\% \text{ of } 6400 + 3\frac{1}{8}\% \text{ of } 9600 + 11.11\% \text{ of } 999 = ? \times 11$$

$$7\frac{1}{2}\% = \frac{3}{40}, 3\frac{1}{8}\% = \frac{1}{32}, \text{ and } 11.11\% = \frac{1}{9}$$

$$\text{So, } \frac{3}{40} \times 6400 + \frac{1}{32} \times 9600 + \frac{1}{9} \times 999 = ? \times 11$$

$$480 + 300 + 111 = ? \times 11$$

$$891 = ? \times 11$$

$$? = 81$$

$$13. \sqrt{?} - 25\% \text{ of } 128 = 58.33\% \text{ of } 432 - 16^2$$

$$25\% \text{ of } 128 = 32 \text{ and } 58.33\% \text{ of } 432 = 252$$

$$\text{So, } \sqrt{?} - 32 = 252 - 256$$

$$\sqrt{?} - 32 = -4$$

$$\sqrt{?} = 28$$

$$? = 784$$

$$14. \sqrt{(64 \times 36)} + \sqrt[3]{(125 \times 216)} + 43.75\% \text{ of } 512 = ? + 91$$

$$\sqrt{(64 \times 36)} = 48, \sqrt[3]{(125 \times 216)} = 30, \text{ and } 43.75\% \text{ of } 512 = 224$$

$$\text{So, } 48 + 30 + 224 = ? + 91$$

$$302 = ? + 91$$

$$? = 211$$

$$15. 12 \times \sqrt{1369} - 16 \times \sqrt{625} + 23 \times \sqrt{196} = ? \times 6$$

$$\sqrt{1369} = 37, \sqrt{625} = 25, \text{ and } \sqrt{196} = 14$$

$$\text{So, } 12 \times 37 - 16 \times 25 + 23 \times 14 = ? \times 6$$

$$444 - 400 + 322 = ? \times 6$$

$$366 = ? \times 6$$

$$? = 61$$

$$16. ?\% \text{ of } 960 + 125\% \text{ of } 480 = 225\% \text{ of } 640 - 360$$

$$125\% \text{ of } 480 = 600 \text{ and } 225\% \text{ of } 640 = 1440$$

$$\text{So, } ?\% \text{ of } 960 + 600 = 1440 - 360$$

$$?\% \text{ of } 960 + 600 = 1080$$

$$?\% \text{ of } 960 = 480$$

$$? = 480 \times 100 \div 960 = 50$$

$$17. [(52^2 - 28^2) \div 16] + \frac{5}{6} \text{ of } 504 - 32.5\% \text{ of } 400 = ? + 77$$

$$52^2 - 28^2 = 2704 - 784 = 1920, \frac{5}{6} \text{ of } 504 = 420, \text{ and } 32.5\% \text{ of } 400 = 130$$

$$\text{So, } 1920 \div 16 + 420 - 130 = ? + 77$$

$$120 + 420 - 130 = ? + 77$$

$$410 = ? + 77$$

$$? = 333$$

$$18. 9^2 + 8^3 - 6.25\% \text{ of } 1024 = (? \div 4) + 17^2$$

$$9^2 = 81, 8^3 = 512, 6.25\% \text{ of } 1024 = 64, \text{ and } 17^2 = 289$$

$$\text{So, } 81 + 512 - 64 = (? \div 4) + 289$$

$$529 = (? \div 4) + 289$$

$$? \div 4 = 240$$

$$? = 960$$

$$19. 41.66\% \text{ of } 1728 - 33.33\% \text{ of } 1080 + 14.28\% \text{ of } 2450 = ? \times 5$$

$$41.66\% = \frac{5}{12}, 33.33\% = \frac{1}{3}, \text{ and } 14.28\% = \frac{1}{7}$$

$$\text{So, } \frac{5}{12} \times 1728 - \frac{1}{3} \times 1080 + \frac{1}{7} \times 2450 = ? \times 5$$

$$720 - 360 + 350 = ? \times 5$$

$$710 = ? \times 5$$

$$? = 142$$

$$20. (3\frac{1}{4} + 2\frac{3}{5}) \times 80 - 68.75\% \text{ of } 640 = ?$$

$$3\frac{1}{4} = \frac{13}{4} \text{ and } 2\frac{3}{5} = \frac{13}{5}$$

$$\text{So, } (\frac{13}{4} + \frac{13}{5}) \times 80 - 68.75\% \text{ of } 640 = ?$$

$$(\frac{65}{20} + \frac{52}{20}) \times 80 - 440 = ?$$

$$\frac{117}{20} \times 80 - 440 = ?$$

$$468 - 440 = ?$$

$$? = 28$$

2. ARITHMETIC QUESTIONS

Q1: Present age of Kartik is $\frac{2}{3}$ of Varun's age 6 years ago. The difference between the present ages of Kartik and Varun is 24 years. If the average present age of Kartik, Varun and Sahil is 42 years, then find Sahil's age 3 years hence.
 कार्तिक की वर्तमान आयु, वरुण की 6 वर्ष पहले की आयु की $\frac{2}{3}$ है। कार्तिक और वरुण की वर्तमान आयु के बीच का अंतर 24 वर्ष है। यदि कार्तिक, वरुण और साहिल की वर्तमान आयु का औसत 42 वर्ष है, तो 3 वर्ष बाद साहिल की आयु ज्ञात कीजिए।

- (A) 29 years
- (B) 35 years
- (C) 39 years
- (D) 33 years
- (E) 31 years

Q2: The volume of a cube whose total surface area is 600 sq cm is 136 cubic cm more than the volume of a sphere. Find the surface area of the sphere. Take $\pi = 3$.

एक घन का आयतन, जिसका कुल पृष्ठीय क्षेत्रफल 600 वर्ग सेमी है, एक गोले के आयतन से 136 घन सेमी अधिक है। गोले का पृष्ठीय क्षेत्रफल ज्ञात कीजिए। $\pi = 3$ लीजिए।

- (A) 384 sq cm
- (B) 432 sq cm
- (C) 450 sq cm
- (D) 576 sq cm
- (E) 500 sq cm

Q3: A cyclist covers half of a journey at a speed of s km/h and the remaining half at 45 km/h. If the total distance is 360 km and the total time taken is 10 hours, then find the value of s .

एक साइकिल चालक यात्रा की आधी दूरी s किमी/घंटा की गति से और शेष आधी दूरी 45 किमी/घंटा की गति से तय करता है। यदि कुल दूरी 360 किमी है और कुल समय 10 घंटे है, तो s का मान ज्ञात कीजिए।

- (A) 25
- (B) 36
- (C) 20
- (D) 40

(E) 30

Q4: A person invests some money in a scheme offering 20% compound interest per annum, compounded half-yearly, for 2 years. At the end of 2 years, he receives Rs. 146410. Find the simple interest earned if the same principal is invested at 12% per annum for 3 years.

एक व्यक्ति कुछ धनराशि ऐसी योजना में निवेश करता है जो 2 वर्षों के लिए 20% वार्षिक चक्रवृद्धि ब्याज देती है, जिसका ब्याज अर्धवार्षिक रूप से जुड़ा है। 2 वर्षों के अंत में उसे 146410 रुपये प्राप्त होते हैं। यदि वही मूलधन 3 वर्षों के लिए 12% वार्षिक साधारण ब्याज पर निवेश किया जाए, तो प्राप्त साधारण ब्याज ज्ञात कीजिए।

- (A) Rs. 30000
- (B) Rs. 42000
- (C) Rs. 36000
- (D) Rs. 32000
- (E) Rs. 40000

Q5: A shopkeeper marked an article 75% above its cost price and allowed two successive discounts of 20% and 20%. He earned a profit of Rs. 96 in the transaction. Find his profit if he had allowed only one discount of 25%.

एक दुकानदार ने एक वस्तु का अंकित मूल्य उसके क्रय मूल्य से 75% अधिक रखा और 20% तथा 20% की दो क्रमागत छूट दी। इस लेन-देन में उसे 96 रुपये का लाभ हुआ। यदि उसने केवल 25% की एक छूट दी होती, तो उसका लाभ ज्ञात कीजिए।

- (A) Rs. 250
- (B) Rs. 180
- (C) Rs. 300
- (D) Rs. 200
- (E) Rs. 225

Q6: The length of train M is x meters. Train N is 120 meters longer than train M. Train M takes the same time to cross a 180-meter-long platform as train N takes to cross a 300-meter-long platform. If the ratio of the speeds of train M and train N is 3:5, then find x .

ट्रेन M की लंबाई x मीटर है। ट्रेन N, ट्रेन M से 120 मीटर अधिक लंबी है। ट्रेन M को 180 मीटर लंबे प्लेटफॉर्म को पार करने में उतना ही समय लगता है जितना ट्रेन N को 300 मीटर लंबे प्लेटफॉर्म को पार करने में लगता है। यदि ट्रेन M और ट्रेन N की गति का अनुपात 3:5 है, तो x ज्ञात कीजिए।

- (A) 150

- (B) 180
- (C) 210
- (D) 240
- (E) 120

Q7: The smallest angle of a triangle is equal to $\frac{1}{4}$ of the largest angle of a quadrilateral. The angles of the quadrilateral are in the ratio 2:3:4:6. The largest angle of the triangle is $\frac{5}{2}$ times its smallest angle. Find the difference between the second largest angle of the triangle and the smallest angle of the quadrilateral.

एक त्रिभुज का सबसे छोटा कोण, एक चतुर्भुज के सबसे बड़े कोण का $\frac{1}{4}$ है। चतुर्भुज के कोणों का अनुपात 2:3:4:6 है। त्रिभुज का सबसे बड़ा कोण उसके सबसे छोटे कोण का $\frac{5}{2}$ गुना है। त्रिभुज के दूसरे सबसे बड़े कोण और चतुर्भुज के सबसे छोटे कोण के बीच का अंतर ज्ञात कीजिए।

- (A) 4°
- (B) 6°
- (C) 8°
- (D) 10°
- (E) 12°

Q8: A box contains green and yellow balls in the ratio 2:3. When 6 green balls are added to the box, the probability of drawing a green ball becomes $\frac{1}{2}$. After this, one ball is removed randomly from the box. Find the probability, in percent, that the maximum number of green balls are left in the box.

एक डिब्बे में हरी और पीली गेंदों का अनुपात 2:3 है। जब डिब्बे में 6 हरी गेंदें डाली जाती हैं, तो हरी गेंद निकालने की प्रायिकता $\frac{1}{2}$ हो जाती है। इसके बाद डिब्बे से एक गेंद यादृच्छिक रूप से निकाली जाती है। डिब्बे में हरी गेंदों की अधिकतम संख्या बची रहने की प्रायिकता प्रतिशत में ज्ञात कीजिए।

- (A) 40%
- (B) 45%
- (C) 50%
- (D) 55%
- (E) 60%

Q9: Two solutions, R and S, contain alcohol and water. Solution R contains 74% alcohol, while solution S contains 38% alcohol. They are mixed in such a ratio that the final mixture contains 54% alcohol. Find by what percentage the quantity of

solution S is more than solution R in the final mixture.

दो विलयन R और S में अल्कोहल और पानी है। विलयन R में 74% अल्कोहल है, जबकि विलयन S में 38% अल्कोहल है। इन्हें ऐसे अनुपात में मिलाया जाता है कि अंतिम मिश्रण में 54% अल्कोहल हो जाता है। अंतिम मिश्रण में विलयन S की मात्रा, विलयन R से कितने प्रतिशत अधिक है?

- (A) 20%
- (B) 25%
- (C) 30%
- (D) 40%
- (E) 50%

Q10: The cost price of an article is Rs. 8000, and it is marked $a\%$ above its cost price. A shopkeeper allows two successive discounts of Rs. 300 and 10% , respectively, and sells it at a profit of Rs. 1450. The cost price of another article is 25% less than that of the first article. If the second article is sold for Rs. 7800 at a profit of $(a - b)\%$, then find b .

एक वस्तु का क्रय मूल्य Rs. 8000 है और इसे इसके क्रय मूल्य से $a\%$ अधिक अंकित किया गया है। एक दुकानदार क्रमशः Rs. 300 और 10% की दो क्रमागत छूट देता है और इसे Rs. 1450 के लाभ पर बेचता है। दूसरी वस्तु का क्रय मूल्य पहली वस्तु से 25% कम है। यदि दूसरी वस्तु Rs. 7800 में $(a - b)\%$ लाभ पर बेची जाती है, तो b ज्ञात कीजिए।

- (A) 3
- (B) 4
- (C) 5
- (D) 6
- (E) 7

Q11: Runner A can cover 1 km in 12 minutes, while Runner B can cover 1 km in 16 minutes. If both run a race and Runner A beats Runner B by 250 meters, find the length of the race.

धावक A 1 किमी की दूरी 12 मिनट में तय कर सकता है, जबकि धावक B 1 किमी की दूरी 16 मिनट में तय कर सकता है। यदि दोनों एक दौड़ में भाग लेते हैं और धावक A, धावक B को 250 मीटर से हरा देता है, तो दौड़ की लंबाई ज्ञात कीजिए।

- (A) 750 m
- (B) 1200 m
- (C) 900 m
- (D) 1000 m

(E) 800 m

Q12: A alone can finish a work in 84 days. B is 50% more efficient than A. A and C together can finish the same work in 42 days. If A, B and C work together, find the time taken to complete the work.

A अकेले किसी कार्य को 84 दिनों में पूरा कर सकता है। B, A से 50% अधिक कुशल है। A और C मिलकर उसी कार्य को 42 दिनों में पूरा कर सकते हैं। यदि A, B और C एक साथ कार्य करते हैं, तो कार्य पूरा करने में लगा समय ज्ञात कीजिए।

(A) 28 days

(B) 24 days

(C) 21 days

(D) 30 days

(E) 36 days

Q13: The cost price of 12 articles is equal to the marked price of 9 articles. If an article is sold at a 25% discount, its selling price is Rs 90 less than its selling price when sold at a 15% discount. Find the cost price of one article.

12 वस्तुओं का क्रय मूल्य 9 वस्तुओं के अंकित मूल्य के बराबर है। यदि एक वस्तु को 25% छूट पर बेचा जाता है, तो उसका विक्रय मूल्य 15% छूट पर बेचने की तुलना में Rs 90 कम होता है। एक वस्तु का क्रय मूल्य ज्ञात कीजिए।

(A) Rs 720

(B) Rs 675

(C) Rs 600

(D) Rs 750

(E) Rs 810

Q14: A solid metal sphere is melted and recast into 96 identical solid cylinders. The radius of each cylinder is 25% of the radius of the sphere, and the height of each cylinder is 2 cm less than its radius. Find the radius of the sphere.

एक ठोस धातु के गोले को पिघलाकर 96 समान ठोस बेलनों में ढाला गया। प्रत्येक बेलन की त्रिज्या, गोले की त्रिज्या का 25% है और प्रत्येक बेलन की ऊंचाई उसकी त्रिज्या से 2 सेमी कम है। गोले की त्रिज्या ज्ञात कीजिए।

(A) 48 cm

(B) 56 cm

(C) 64 cm

(D) 80 cm

(E) 72 cm

Q15: One customer bought 4 kg of rice and 5 kg of lentils for Rs 465. Another customer bought 3 kg of rice and 8 kg of lentils for Rs 540. Find the price per kg of lentils as a percentage of the price per kg of rice.

एक ग्राहक ने 4 किग्रा चावल और 5 किग्रा दाल Rs 465 में खरीदी। दूसरे ग्राहक ने 3 किग्रा चावल और 8 किग्रा दाल Rs 540 में खरीदी। दाल का प्रति किग्रा मूल्य, चावल के प्रति किग्रा मूल्य का कितना प्रतिशत है, ज्ञात कीजिए।

- (A) 75%
- (B) 60%
- (C) 80%
- (D) 90%
- (E) 70%

Q16: A person borrowed Rs. 30000 at 20% p.a. compound interest. After one year, he borrowed Rs. X more at the same rate of interest. At the end of the third year, he paid a total amount of Rs. 59040. Find the value of X.

एक व्यक्ति ने 20% वार्षिक चक्रवृद्धि ब्याज पर 30000 रुपये उधार लिए। एक वर्ष बाद उसने उसी ब्याज दर पर x रुपये और उधार लिए। तीसरे वर्ष के अंत में उसने कुल 59040 रुपये चुकाए। x का मान ज्ञात कीजिए।

- (A) Rs. 4000
- (B) Rs. 6000
- (C) Rs. 4500
- (D) Rs. 5000
- (E) Rs. 7000

Q17: Two persons invested Rs. X and Rs. Y in the same simple interest scheme at 12% p.a. for 3 years. The interest received on Rs. X is Rs. 864 more than the interest received on Rs. Y. If X is 40% more than Y, find the total amount invested by them together.

दो व्यक्तियों ने समान साधारण ब्याज योजना में क्रमशः x रुपये और y रुपये 12% वार्षिक दर से 3 वर्षों के लिए निवेश किए। x रुपये पर प्राप्त ब्याज, y रुपये पर प्राप्त ब्याज से 864 रुपये अधिक है। यदि x, y से 40% अधिक है, तो दोनों द्वारा मिलाकर निवेश की गई कुल राशि ज्ञात कीजिए।

- (A) Rs. 12600
- (B) Rs. 14400
- (C) Rs. 15000

(D) Rs. 13200

(E) Rs. 15600

Q18: A garment seller marked a jacket 60% above its cost price and then allowed two successive discounts of $x\%$ and $(x + 20)\%$. If he incurred a loss of 34% on selling the jacket, find the value of x .

एक वस्त्र विक्रेता ने एक जैकेट का अंकित मूल्य उसके क्रय मूल्य से 60% अधिक रखा और फिर $x\%$ तथा $(x + 20)\%$ की दो क्रमिक छूट दी। यदि जैकेट बेचने पर उसे 34% की हानि हुई, तो x का मान ज्ञात कीजिए।

(A) 35

(B) 20

(C) 25

(D) 15

(E) 30

Q19: An amount is distributed among four persons K, L, M and N. The amounts of K, L and M are in the ratio 5:7:8. Person N receives an amount equal to half of the combined amount of K, L and M. If the difference between the average amount of K and M and the average amount of L and N is Rs. 600, find the total amount distributed among all four persons.

एक राशि चार व्यक्तियों K, L, M और N में बाँटी गई। K, L और M की राशियों का अनुपात 5:7:8 है। N को K, L और M की कुल राशि के आधे के बराबर राशि मिलती है। यदि K और M की औसत राशि तथा L और N की औसत राशि के बीच का अंतर 600 रुपये है, तो चारों व्यक्तियों में बाँटी गई कुल राशि ज्ञात कीजिए।

(A) Rs. 9000

(B) Rs. 7800

(C) Rs. 8400

(D) Rs. 9600

(E) Rs. 10200

Q20: When the digits of a two-digit number are reversed, the number obtained is 36 less than twice the original number. If the reversed number is 120% of the original number, find the sum of the digits of the original number.

एक दो अंकों की संख्या के अंकों को उलटने पर प्राप्त संख्या, मूल संख्या के दोगुने से 36 कम है। यदि उलटी संख्या, मूल संख्या का 120% है, तो मूल संख्या के अंकों का योग ज्ञात कीजिए।

(A) 12

- (B) 10
(C) 13
(D) 8
(E) 9

Answers:

- | | |
|------|------|
| 1. D | 11.D |
| 2. B | 12.B |
| 3. E | 13.B |
| 4. C | 14.E |
| 5. A | 15.A |
| 6. B | 16.D |
| 7. B | 17.B |
| 8. C | 18.C |
| 9. B | 19.A |
| 10.C | 20.E |

Solutions:**1. Option D**

Let Varun's present age be V years.

Kartik's present age = $\frac{2}{3}$ of Varun's age 6 years ago

Kartik's present age = $\frac{2}{3} \times (V - 6)$

Given difference between Varun and Kartik = 24 years

$$V - \frac{2}{3} \times (V - 6) = 24$$

$$3V - 2V + 12 = 72$$

$$V = 60$$

Varun's present age = 60 years

Kartik's present age = $60 - 24 = 36$ years

Average present age of Kartik, Varun and Sahil = 42 years

Total present age = $42 \times 3 = 126$ years

Sahil's present age = $126 - 60 - 36 = 30$ years

Sahil's age 3 years hence = $30 + 3 = 33$ years

2. Option B

Total surface area of cube = $6 \times \text{side} \times \text{side}$

$$600 = 6 \times \text{side} \times \text{side}$$

$$\text{Side} \times \text{side} = 600/6 = 100$$

$$\text{Side} = 10 \text{ cm}$$

$$\text{Volume of cube} = 10 \times 10 \times 10 = 1000 \text{ cubic cm}$$

$$\text{Volume of sphere} = 1000 - 136 = 864 \text{ cubic cm}$$

$$\text{Take } \pi = 3$$

$$\text{Volume of sphere} = \frac{4}{3} \times \pi \times r \times r \times r$$

$$864 = \frac{4}{3} \times 3 \times r \times r \times r$$

$$864 = 4 \times r \times r \times r$$

$$r \times r \times r = 864/4 = 216$$

$$r = 6 \text{ cm}$$

$$\text{Surface area of sphere} = 4 \times \pi \times r \times r$$

$$\text{Surface area} = 4 \times 3 \times 6 \times 6 = 432 \text{ sq cm}$$

3. Option E

$$\text{Total distance} = 360 \text{ km}$$

$$\text{Half distance} = 360/2 = 180 \text{ km}$$

$$\text{Time taken for remaining half at } 45 \text{ km/h} = 180/45 = 4 \text{ hours}$$

$$\text{Total time} = 10 \text{ hours}$$

$$\text{Time taken for first half} = 10 - 4 = 6 \text{ hours}$$

$$\text{Speed for first half} = \text{Distance/Time}$$

$$s = 180/6 = 30 \text{ km/h}$$

4. Option C

$$\text{Rate} = 20\% \text{ per annum compounded half-yearly}$$

$$\text{Half-yearly rate} = 20/2 = 10\%$$

$$\text{Time} = 2 \text{ years}$$

$$\text{Number of half-years} = 2 \times 2 = 4$$

$$\text{Amount} = \text{Principal} \times 110/100 \times 110/100 \times 110/100 \times 110/100$$

$$146410 = \text{Principal} \times 14641/10000$$

$$\text{Principal} = 146410 \times 10000/14641$$

$$\text{Principal} = \text{Rs. } 100000$$

$$\text{Simple interest} = \text{Principal} \times \text{Rate} \times \text{Time}/100$$

$$\text{Simple interest} = 100000 \times 12 \times 3/100$$

$$\text{Simple interest} = \text{Rs. } 36000$$

5. Option A

Let the cost price be 100 units.

Marked price = 175% of cost price = 175 units

After first 20% discount:

Price = $175 \times 80/100 = 140$ units

After second 20% discount:

Selling price = $140 \times 80/100 = 112$ units

Profit = $112 - 100 = 12$ units

Given profit = Rs. 96

12 units = 96

1 unit = $96/12 = \text{Rs. } 8$

Cost price = $100 \times 8 = \text{Rs. } 800$

Marked price = $175 \times 8 = \text{Rs. } 1400$

If only one discount of 25% is given:

Selling price = $1400 \times 75/100 = \text{Rs. } 1050$

Profit = $1050 - 800 = \text{Rs. } 250$

6. Option B

Let the length of train M = x meters.

Length of train N = $x + 120$ meters.

Train M crosses a 180-meter platform, so total distance covered = $x + 180$ meters.

Train N crosses a 300-meter platform, so total distance covered = $x + 120 + 300 = x + 420$ meters.

Ratio of speeds of train M and train N = 3:5.

Since time taken is the same, distance ratio will be equal to speed ratio.

So, $(x + 180)/(x + 420) = 3/5$

$5(x + 180) = 3(x + 420)$

$5x + 900 = 3x + 1260$

$2x = 360$

$x = 180$ meters

7. Option B

Sum of angles of a quadrilateral = 360° .

Ratio of angles = 2:3:4:6

Total parts = $2 + 3 + 4 + 6 = 15$

1 part = $360/15 = 24^\circ$

Angles of quadrilateral are 48° , 72° , 96° and 144° .

Largest angle of quadrilateral = 144°

Smallest angle of triangle = $1/4$ of $144 = 36^\circ$

Largest angle of triangle = $5/2 \times 36 = 90^\circ$

Sum of angles of triangle = 180°

Second largest angle of triangle = $180 - 90 - 36 = 54^\circ$

Smallest angle of quadrilateral = 48°

Required difference = $54 - 48 = 6^\circ$

8. Option C

Let the number of green and yellow balls be $2x$ and $3x$.

After adding 6 green balls, green balls = $2x + 6$

Total balls = $2x + 6 + 3x = 5x + 6$

Probability of drawing a green ball = $1/2$

So, $(2x + 6)/(5x + 6) = 1/2$

$2(2x + 6) = 5x + 6$

$4x + 12 = 5x + 6$

$x = 6$

Initial green balls = $2 \times 6 = 12$

Initial yellow balls = $3 \times 6 = 18$

After adding 6 green balls, green balls = $12 + 6 = 18$

Yellow balls = 18

Total balls = 36

For the maximum number of green balls to be left, the removed ball must be yellow.

Required probability = $18/36 = 1/2 = 50\%$

9. Option B

Solution R contains 74% alcohol.

Solution S contains 38% alcohol.

Final mixture contains 54% alcohol.

Using alligation:

Quantity of R : Quantity of S = $(54 - 38) : (74 - 54)$

Quantity of R : Quantity of S = $16 : 20 = 4 : 5$

So, solution R = 4 parts and solution S = 5 parts.

Difference = $5 - 4 = 1$ part

Percentage by which S is more than R = $1/4 \times 100 = 25\%$

10. Option C

Cost price of first article = Rs. 8000

Marked price is $a\%$ above cost price.

After a discount of Rs. 300 and then 10%, the article is sold at a profit of Rs. 1450.

Selling price of first article = $8000 + 1450 = \text{Rs. } 9450$

After 10% discount, selling price is 90% of the price after Rs. 300 discount.

So, price after Rs. 300 discount = $9450 \times 100/90 = \text{Rs. } 10500$

Marked price = $10500 + 300 = \text{Rs. } 10800$

Markup = $10800 - 8000 = \text{Rs. } 2800$

$a\% = 2800/8000 \times 100 = 35\%$

Cost price of second article = 25% less than Rs. 8000

Cost price of second article = $75/100 \times 8000 = \text{Rs. } 6000$

Selling price of second article = Rs. 7800

Profit = $7800 - 6000 = \text{Rs. } 1800$

Profit percent = $1800/6000 \times 100 = 30\%$

Given profit percent = $(a - b)\%$

So, $a - b = 30$

$35 - b = 30$

$b = 5$

11. Option D

Time taken to cover 1 km by Runner A = 12 minutes.

Time taken to cover 1 km by Runner B = 16 minutes.

Speed ratio of Runner A and Runner B = $16:12 = 4:3$.

When Runner A completes the race, Runner B covers $3/4$ of the race.

Difference covered = $1/4$ of the race.

Given difference = 250 m.

So, $1/4$ of race length = 250.

Race length = $250 \times 4 = 1000$ m.

12. Option B

Take total work = 84 units.

A alone completes the work in 84 days, so A's efficiency = $84 \div 84 = 1$ unit/day.

B is 50% more efficient than A.

So, B's efficiency = $1 + 50\% \text{ of } 1 = 1.5$ units/day.

A and C together complete the work in 42 days.

Efficiency of A and C together = $84 \div 42 = 2$ units/day.

C's efficiency = $2 - 1 = 1$ unit/day.

Total efficiency of A, B and C = $1 + 1.5 + 1 = 3.5$ units/day.

Required time = $84 \div 3.5 = 24$ days.

13. Option B

Cost price of 12 articles = Marked price of 9 articles.

So, $12 \times CP = 9 \times MP$.

$MP = 12CP/9 = 4CP/3$.

Difference between 25% discount and 15% discount = 10% of MP.

Given, 10% of MP = Rs 90.

So, $MP = 90 \times 100/10 = \text{Rs } 900$.

Now, $900 = 4CP/3$.

$CP = 900 \times 3/4 = \text{Rs } 675$.

14. Option E

Let the radius of the sphere be R cm.

Radius of each cylinder = 25% of R = $R/4$.

Height of each cylinder = $R/4 - 2$.

Volume of sphere = Total volume of 96 cylinders.

Using volume shortcut:

$\frac{4}{3} \times R \times R \times R = 96 \times R/4 \times R/4 \times (R/4 - 2)$.

$4R/3 = 96/16 \times (R/4 - 2)$.

$4R/3 = 6 \times (R/4 - 2)$.

$4R/3 = 3R/2 - 12$.

Multiplying by 6:

$8R = 9R - 72$.

$R = 72$ cm.

15. Option A

Let the price of rice be Rs R per kg and the price of lentils be Rs L per kg.

According to the first condition:

$4R + 5L = 465$.

According to the second condition:

$3R + 8L = 540$.

Multiply the first equation by 3:

$12R + 15L = 1395$.

Multiply the second equation by 4:

$$12R + 32L = 2160.$$

Subtract:

$$17L = 765.$$

$$L = 45.$$

$$\text{Now, } 4R + 5 \times 45 = 465.$$

$$4R + 225 = 465.$$

$$4R = 240.$$

$$R = 60.$$

$$\text{Required percentage} = 45 \times 100/60 = 75\%.$$

16. Option D

First amount borrowed = Rs. 30000

Rate = 20% p.a.

Time for first borrowing = 3 years

$$\text{Amount of first borrowing after 3 years} = 30000 \times 120/100 \times 120/100 \times 120/100$$

$$\text{Amount of first borrowing} = 30000 \times 216/125$$

$$\text{Amount of first borrowing} = \text{Rs. } 51840$$

$$\text{Total amount paid} = \text{Rs. } 59040$$

$$\text{Amount paid for second borrowing} = 59040 - 51840 = \text{Rs. } 7200$$

Second borrowing was taken after one year, so it remained for 2 years.

$$\text{Amount of Rs. } X \text{ after 2 years} = X \times 120/100 \times 120/100$$

$$7200 = X \times 144/100$$

$$X = 7200 \times 100/144$$

$$X = 5000$$

17. Option B

X is 40% more than Y.

So, take Y = 100 parts and X = 140 parts.

Difference between X and Y = 40 parts

Rate = 12% p.a.

Time = 3 years

Interest difference = Rs. 864

Simple interest on the difference of principal = 864

$$40 \text{ parts} \times 12 \times 3/100 = 864$$

$$40 \text{ parts} \times 36/100 = 864$$

$$14.4 \text{ parts} = 864$$

$$1 \text{ part} = 864/14.4 = 60$$

$$Y = 100 \text{ parts} = 100 \times 60 = \text{Rs. } 6000$$

$$X = 140 \text{ parts} = 140 \times 60 = \text{Rs. } 8400$$

$$\text{Total amount invested} = 6000 + 8400 = \text{Rs. } 14400$$

18. Option C

$$\text{Let cost price} = \text{Rs. } 100$$

Marked price is 60% above cost price.

$$\text{Marked price} = 100 + 60 = \text{Rs. } 160$$

$$\text{Loss} = 34\%$$

$$\text{Selling price} = 100 - 34 = \text{Rs. } 66$$

Now, marked price after two successive discounts becomes selling price.

$$160 \times (100 - x)/100 \times (100 - x - 20)/100 = 66$$

$$160 \times (100 - x)/100 \times (80 - x)/100 = 66$$

Using options, check $x = 25$.

$$\text{First discount} = 25\%$$

$$\text{Second discount} = 25 + 20 = 45\%$$

$$\text{Selling price} = 160 \times 75/100 \times 55/100$$

$$\text{Selling price} = 160 \times 3/4 \times 11/20$$

$$\text{Selling price} = 66$$

This matches the required selling price.

Therefore, $x = 25$.

19. Option A

Let the amounts of K, L and M be $5a$, $7a$ and $8a$.

$$\text{Combined amount of K, L and M} = 5a + 7a + 8a = 20a$$

$$\text{Amount of N} = \text{half of } 20a = 10a$$

$$\text{Average amount of K and M} = (5a + 8a)/2 = 13a/2$$

$$\text{Average amount of L and N} = (7a + 10a)/2 = 17a/2$$

$$\text{Difference between both averages} = 17a/2 - 13a/2 = 4a/2 = 2a$$

$$\text{Given difference} = \text{Rs. } 600$$

$$2a = 600$$

$$a = 300$$

$$\text{Total amount distributed} = 5a + 7a + 8a + 10a$$

$$\text{Total amount distributed} = 30a$$

$$\text{Total amount distributed} = 30 \times 300 = \text{Rs. } 9000$$

20. Option E

Reversed number is 120% of the original number.

$$120\% = 120/100 = 6/5$$

So, original number : reversed number = 5 : 6

Let original number = $5a$

Reversed number = $6a$

It is also given that the reversed number is 36 less than twice the original number.

So, $2 \times \text{original number} - \text{reversed number} = 36$

$$2 \times 5a - 6a = 36$$

$$10a - 6a = 36$$

$$4a = 36$$

$$a = 9$$

Original number = $5a = 5 \times 9 = 45$

Reversed number = $6a = 6 \times 9 = 54$

Sum of digits of the original number = $4 + 5 = 9$

Therefore, the sum of the digits is 9.

3. Quadratic Equations

CHECKLIST

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

A. if $x > y$

B. if $x \geq y$

C. if $x < y$

D. if $x \leq y$

E. if $x = y$ or relationship cannot be established

1.) I. $4x^2 - 49x + 111 = 0$

II. $12y^2 - 35y + 8 = 0$

2.) I. $18x^2 - 90x + 88 = 0$

II. $5y^2 - 56y + 156 = 0$

3.) I. $8x^2 - 38x + 17 = 0$

II. $4y^2 - 57y + 170 = 0$

4.) I. $x^2 - 8\sqrt{5}x + 35 = 0$

II. $y^2 - 8\sqrt{5}y + 60 = 0$

5.) I. $4x^2 - 43x + 105 = 0$

II. $8y^2 - 54y + 90 = 0$

6.) I. $3x^2 - 40x + 132 = 0$

II. $15y^2 - 93y + 132 = 0$

7.) I. $18x^2 - 21x + 6 = 0$

II. $12y^2 - 97y + 133 = 0$

8.) I. $2x^2 - 35x + 152 = 0$

II. $3y^2 - 34y + 80 = 0$

9.) I. $10x^2 - 28x + 16 = 0$

II. $5y^2 - 34y + 48 = 0$

10.) I. $3x^2 - 33x + 90 = 0$

II. $10y^2 - 39y + 17 = 0$

11.) I. $x^2 - 5\sqrt{13}x + 78 = 0$

II. $y^2 - 6\sqrt{13}y + 65 = 0$

12.) I. $10x^2 - 87x + 143 = 0$

II. $15y^2 - 43y + 22 = 0$

13.) I. $15x^2 - 26x + 7 = 0$

II. $8y^2 - 70y + 132 = 0$

14.) I. $2x^2 - 36x + 160 = 0$

II. $10y^2 - 84y + 170 = 0$

15.) I. $5x^2 - 34x + 45 = 0$

II. $4y^2 - 55y + 175 = 0$

16.) I. $x^2 - 4\sqrt{14}x + 42 = 0$

II. $y^2 - 4\sqrt{14}y + 56 = 0$

17.) I. $18x^2 - 15x + 3 = 0$

II. $4y^2 - 45y + 74 = 0$

18.) I. $2x^2 - 16x + 14 = 0$

II. $16y^2 - 20y + 4 = 0$

19.) I. $5x^2 - 52x + 119 = 0$

II. $15y^2 - 23y + 6 = 0$

20.) I. $8x^2 - 10x + 3 = 0$

II. $16y^2 - 44y + 24 = 0$

Answers:

1. A

2. C

3. D

4. E

5. B

6. A

7. C

8. B

9. D

10. A

11. E

12. B

13. C

14. A

15. D

16. E

17. C

18. B

19. A

20. D

Solutions:

$$(1) x = 3, \frac{37}{4}$$

$$y = \frac{1}{4}, \frac{8}{3}$$

$$(2) x = \frac{4}{3}, \frac{11}{3}$$

$$y = \frac{26}{5}, 6$$

$$(3) x = \frac{1}{2}, \frac{17}{4}$$

$$y = \frac{17}{4}, 10$$

$$(4) x = \sqrt{5}, 7\sqrt{5}$$

$$y = 2\sqrt{5}, 6\sqrt{5}$$

$$(5) x = \frac{15}{4}, 7$$

$$y = 3, \frac{15}{4}$$

$$(6) x = 6, \frac{22}{3}$$

$$y = \frac{11}{5}, 4$$

$$(7) x = \frac{1}{2}, \frac{2}{3}$$

$$y = \frac{7}{4}, \frac{19}{3}$$

$$(8) x = 8, \frac{19}{2}$$

$$y = \frac{10}{3}, 8$$

$$(9) x = \frac{4}{5}, 2$$

$$y = 2, \frac{24}{5}$$

$$(10) x = 5, 6$$

$$y = \frac{1}{2}, \frac{17}{5}$$

$$(11) x = 2\sqrt{13}, 3\sqrt{13}$$

$$y = \sqrt{13}, 5\sqrt{13}$$

$$(12) x = \frac{11}{5}, \frac{13}{2}$$

$$y = \frac{2}{3}, \frac{11}{5}$$

$$(13) x = \frac{1}{3}, \frac{7}{5}$$

$$y = \frac{11}{4}, 6$$

$$(14) x = 8, 10$$

$$y = \frac{17}{5}, 5$$

$$(15) x = \frac{9}{5}, 5$$

$$y = 5, \frac{35}{4}$$

$$(16) x = \sqrt{14}, 3\sqrt{14}$$

$$y = 2\sqrt{14}, 2\sqrt{14}$$

$$(17) x = \frac{1}{3}, \frac{1}{2}$$

$$y = 2, \frac{37}{4}$$

$$(18) x = 1, 7$$

$$y = \frac{1}{4}, 1$$

$$(19) x = \frac{17}{5}, 7$$

$$y = \frac{1}{3}, \frac{6}{5}$$

$$(20) x = \frac{1}{2}, \frac{3}{4}$$

$$y = \frac{3}{4}, 2$$

4. WRONG NUMBER SERIES

(1) 2240, 2264, 2294, 2348, 2388, 2460

- (a) 2264
- (b) 2348
- (c) 2388
- (d) 2460
- (e) None of these

(2) 3850, 3818, 3753, 3683, 3596, 3490

- (a) 3818
- (b) 3753
- (c) 3596
- (d) 3490
- (e) None of these

(3) 142, 305, 943, 3747, 18747, 112491

- (a) 305
- (b) 3747
- (c) 18747
- (d) 943
- (e) None of these

(4) 91330, 45690, 15260, 3850, 800, 180

- (a) 45690
- (b) 3850
- (c) 800
- (d) 180
- (e) None of these

(5) 340, 220, 630, 480, 1190, 1000

- (a) 340
- (b) 630
- (c) 480
- (d) 1000
- (e) None of these

(6) 210, 175, 365, 520, 835, 1295

- (a) 365
- (b) 520
- (c) 835
- (d) 1295
- (e) None of these

(7) 760, 783, 829, 932, 1105, 1473

- (a) 783

- | | |
|---|--|
| (b) 829 | (11) 910, 930, 900, 942, 876, 958 |
| (c) 1105 | (a) 942 |
| (d) 932 | (b) 876 |
| (e) None of these | (c) 900 |
| | (d) 930 |
| (8) 2500, 2320, 2240, 2185, 2162.5, 2151.25 | (e) None of these |
| (a) 2162.5 | (12) 300.5, 310, 324.5, 354, 368.5, 398 |
| (b) 2320 | (a) 310 |
| (c) 2240 | (b) 368.5 |
| (d) 2151.25 | (c) 398 |
| (e) None of these | (d) 354 |
| (9) 575, 675, 783, 909, 1023, 1155 | (e) None of these |
| (a) 909 | (13) 315, 641, 1304, 2642, 5392, 10960 |
| (b) 675 | (a) 1304 |
| (c) 1023 | (b) 641 |
| (d) 1155 | (c) 2642 |
| (e) None of these | (d) 10960 |
| (10) 480, 502, 561, 691, 892, 1230 | (e) None of these |
| (a) 502 | (14) 11900, 5970, 2960, 1460, 715, 347.5 |
| (b) 561 | (a) 11900 |
| (c) 892 | (b) 2960 |
| (d) 1230 | (c) 715 |
| (e) None of these | |

(d) 347.5

(e) None of these

(15) 1200, 1209, 1246, 1290, 1380, 1515

(a) 1209

(b) 1290

(c) 1380

(d) 1515

(e) None of these

(16) 160, 464, 1367, 4075, 12146, 36374

(a) 1367

(b) 4075

(c) 12146

(d) 36374

(e) None of these

(17) 600, 628, 673, 739, 840, 950

(a) 628

(b) 673

(c) 950

(d) 840

(e) None of these

(18) 1600, 1580, 1545, 1508, 1452, 1380

(a) 1545

(b) 1508

(c) 1452

(d) 1380

(e) None of these

(19) 125, 500, 250, 1250, 615, 3750

(a) 500

(b) 1250

(c) 615

(d) 3750

(e) None of these

(20) 900, 916, 965, 1075, 1234, 1490

(a) 916

(b) 965

(c) 1234

(d) 1490

(e) None of these

Answers:

(1) b

(2) a

(3) d

(4) c

(5) e

(6) b

(7) d

(8) c

(9) a

(10) e

(11) b

(12) d

(13) c

(14) a

(15) e

(16) b

(17) d

(18) a

(19) c

(20) e

Solutions:

(1) +24, +30, +40, +54, +72; second differences: +6, +10, +14, +18.

(2) -42, -55, -70, -87, -106; second differences: -13, -15, -17, -19.

(3) $\times 2 + 21$, $\times 3 + 18$, $\times 4 + 15$, $\times 5 + 12$, $\times 6 + 9$.

(4) $\div 2 + 25$, $\div 3 + 30$, $\div 4 + 35$, $\div 5 + 40$, $\div 6 + 45$.

(5) Odd positions: $\times 2 - 50$; Even positions: $\times 2 + 40$.

(6) $210 + 175 - 20 = 365$; $175 + 365 - 30 = 510$; $365 + 510 - 40 = 835$; $510 + 835 - 50 = 1295$.

(7) +23, +46, +92, +184, +368; each next difference = previous difference $\times 2$.

(8) -180, -90, -45, -22.5, -11.25; each next difference is half of the previous difference.

(9) 23×25 , 25×27 , 27×29 , 29×31 , 31×33 , 33×35 .

(10) $+(3^3 - 5)$, $+(4^3 - 5)$, $+(5^3 - 5)$, $+(6^3 - 5)$, $+(7^3 - 5)$.

(11) $+(5 \times 4)$, $-(6 \times 5)$, $+(7 \times 6)$, $-(8 \times 7)$, $+(9 \times 8)$.

(12) +9.5, +14.5, +19.5, +24.5, +29.5.

(13) $\times 2 + 11$, $\times 2 + 22$, $\times 2 + 44$, $\times 2 + 88$, $\times 2 + 176$.

(14) $\div 2 - 30$, $\div 2 - 25$, $\div 2 - 20$, $\div 2 - 15$, $\div 2 - 10$.

(15) $+(9 \times 1)$, $+(9 \times 3)$, $+(9 \times 6)$, $+(9 \times 10)$, $+(9 \times 15)$.

(16) $\times 3 - 4^2$, $\times 3 - 5^2$, $\times 3 - 6^2$, $\times 3 - 7^2$, $\times 3 - 8^2$.

(17) $+(4 \times 7)$, $+(5 \times 9)$, $+(6 \times 11)$, $+(7 \times 13)$, $+(8 \times 15)$.

(18) $-(4^2 + 4)$, $-(5^2 + 5)$, $-(6^2 + 6)$, $-(7^2 + 7)$, $-(8^2 + 8)$.

(19) $\times 4$, $\div 2$, $\times 5$, $\div 2$, $\times 6$.

(20) $+4^2$, $+7^2$, $+10^2$, $+13^2$, $+16^2$.

5. MISSING NUMBER SERIES

(1) 24, 33, 53, ?, 134, 199

- (a) 84
- (b) 86
- (c) 88
- (d) 90
- (e) None of these

(2) 8240, 8213, 8159, ?, 7934, 7745

- (a) 8063
- (b) 8065
- (c) 8067
- (d) 8069
- (e) None of these

(3) 6, 16, 57, ?, 1245, 7506

- (a) 244
- (b) 246
- (c) 248
- (d) 250
- (e) None of these

(4) 65, 98, 167, ?, 457, 756

- (a) 270
- (b) 272
- (c) 274
- (d) 276
- (e) None of these

(5) 5184, 1728, 432, ?, 18, 3

- (a) 70
- (b) 72
- (c) 74
- (d) 76
- (e) None of these

(6) 26.5, 31.4, 38.7, ?, 60.5, 75

- (a) 47.8
- (b) 48
- (c) 48.2
- (d) 48.4
- (e) None of these

(7) 2300, 2421, 2565, ?, 2930, 3155

- (a) 2734
- (b) 2738
- (c) 2740
- (d) 2742
- (e) None of these

(8) 54, 105, 202, ?, 764, 1505

- (a) 389
- (b) 391
- (c) 393

(d) 395
 (e) None of these
 (9) 945, 936, 911, ?, 781, 660

(a) 856
 (b) 858
 (c) 860
 (d) 862
 (e) None of these

(10) 6, 20, 50, ?, 182, 296

(a) 102
 (b) 104
 (c) 106
 (d) 108
 (e) None of these

(11) 8974, 4497, 1508, ?, 84, 20

(a) 381
 (b) 383
 (c) 385
 (d) 387
 (e) None of these

(12) 68, 92, 127, 175, ?, 318

(a) 232
 (b) 234
 (c) 236
 (d) 238
 (e) None of these

(13) 980, 968, 947, ?, 870, 810

(a) 909
 (b) 911
 (c) 913
 (d) 915
 (e) None of these

(14) 9, 13, 35, ?, 509, 2581

(a) 117
 (b) 119
 (c) 121
 (d) 123
 (e) None of these

(15) 3560, 3576, 3612, ?, 3776, 3920

(a) 3672
 (b) 3674
 (c) 3676
 (d) 3678
 (e) None of these

(16) 32, 58, 102, ?, 268, 402

(a) 168
 (b) 170
 (c) 172
 (d) 174
 (e) None of these

(17) 7200, 7180, 7120, ?, 6724, 6280

(a) 6984
 (b) 6988
 (c) 6992
 (d) 6996
 (e) None of these

(18) 43, 55, 73, ?, 143, 203

(a) 97
 (b) 99
 (c) 101
 (d) 103
 (e) None of these

(19) 4860, 4872, 4896, ?, 5040, 5232

(a) 4936
 (b) 4944

- (c) 4952
 (d) 4960
 (e) None of these
 (20) 28, 38, 62, ?, 176, 278

- (a) 102
 (b) 104
 (c) 106
 (d) 108
 (e) None of these

Answers:

- (1) b
 (2) d
 (3) a
 (4) c
 (5) b
 (6) d
 (7) a
 (8) b
 (9) d
 (10) a
 (11) c
 (12) d
 (13) d
 (14) c
 (15) c
 (16) b
 (17) a
 (18) c
 (19) b
 (20) c

Solutions:

- (1) +9, +20, +33, +48, +65

Second difference: +11, +13, +15, +17

(2) $-9 \times 3, -9 \times 6, -9 \times 10, -9 \times 15, -9 \times 21$

(3) $\times 2 + 2^2, \times 3 + 3^2, \times 4 + 4^2, \times 5 + 5^2, \times 6 + 6^2$

(4) Sum of previous two numbers
 $+2^2, +3^2, +4^2, +5^2$

(5) $\div 3, \div 4, \div 6, \div 4, \div 6$

(6) +4.9, +7.3, +9.7, +12.1, +14.5

Difference increases by +2.4 each time.

(7) $+11^2, +12^2, +13^2, +14^2, +15^2$

(8) $\times 2 - 3, \times 2 - 8, \times 2 - 13, \times 2 - 18, \times 2 - 23$

(9) $-3^2, -5^2, -7^2, -9^2, -11^2$

(10) Terms are: $n^3 - n^2 + 2$, for $n = 2, 3, 4, 5, 6, 7$

(11) $\div 2 + 10, \div 3 + 9, \div 4 + 8, \div 5 + 7, \div 6 + 6$

(12) $+6 \times 4, +7 \times 5, +8 \times 6, +9 \times 7, +10 \times 8$

(13) -12, -21, -32, -45, -60

Each subtraction = $n^2 + 4n$, for $n = 2, 3, 4, 5, 6$

(14) $\times 1 + 2^2, \times 2 + 3^2, \times 3 + 4^2, \times 4 + 5^2, \times 5 + 6^2$

(15) $+4^2, +6^2, +8^2, +10^2, +12^2$

(16) Terms are: $n^3 + 7n + 10$, for $n = 2, 3, 4, 5, 6, 7$

(17) -20, -60, -136, -260, -444

Each subtraction = $2 \times (n^3 + n)$, for n
= 2, 3, 4, 5, 6

(18) +12, +18, +28, +42, +60

Second difference: +6, +10, +14,
+18

(19) +12, +24, +48, +96, +192

(20) +2×5, +3×8, +4×11, +5×14,
+6×17

CHECKLIST

BY

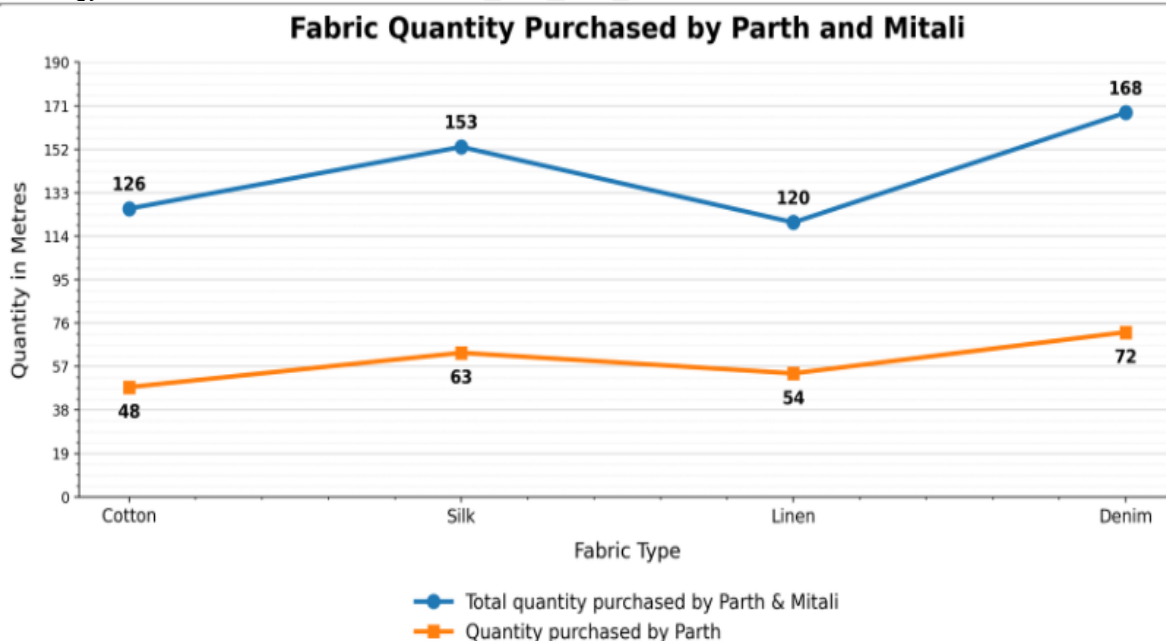
AASHISH

ARORA

6. DATA INTERPRETATION

Set 1: The line graph shows the total quantity of four different fabrics, i.e., Cotton, Silk, Linen and Denim, purchased by two boutique owners, Parth and Mitali, and also shows the quantity of each fabric purchased by Parth. Total quantity of fabric purchased = Quantity purchased by Parth + Quantity purchased by Mitali. Study the line graph carefully and answer the following questions.

रेखा आलेख चार अलग-अलग कपड़ों, अर्थात् कपास, रेशम, लिनन और डेनिम की कुल मात्रा को दर्शाता है, जिन्हें दो बुटीक मालिकों, पार्थ और मिताली ने खरीदा है। यह पार्थ द्वारा खरीदे गए प्रत्येक कपड़े की मात्रा भी दर्शाता है। खरीदे गए कपड़े की कुल मात्रा = पार्थ द्वारा खरीदी गई मात्रा + मिताली द्वारा खरीदी गई मात्रा। रेखा आलेख का ध्यानपूर्वक अध्ययन करें और निम्नलिखित प्रश्नों के उत्तर दें।



Q1: The quantity of cotton fabric purchased by Kabir is $33\frac{1}{3}\%$ more than the quantity of silk fabric purchased by Mitali. The quantity of linen fabric purchased by Kabir is equal to the average quantity of cotton and denim fabrics purchased by Parth. Find the total quantity of cotton and linen fabrics purchased by Kabir.

कबीर द्वारा खरीदे गए कपास के कपड़े की मात्रा, मिताली द्वारा खरीदे गए रेशम के कपड़े की मात्रा से $33\frac{1}{3}\%$ अधिक है। कबीर द्वारा खरीदे गए लिनन के कपड़े की मात्रा, पार्थ द्वारा खरीदे गए कपास और डेनिम के कपड़ों की औसत मात्रा के बराबर है। कबीर द्वारा खरीदे

गए कपास और लिनन के कपड़ों की कुल मात्रा ज्ञात कीजिए।

- (A) 168
- (B) 192
- (C) 174
- (D) 180
- (E) 186

Q2: Mitali purchased two types of silk fabric, Handloom and Printed, in the ratio 4 : 5 respectively. Parth purchased two types of linen fabric, A-type and B-type, and the quantity of B-type linen purchased by him is $66\frac{2}{3}\%$ of the total linen purchased by him. Find the difference between the quantity of Handloom silk purchased by Mitali and A-type linen purchased by Parth.

मिताली ने दो प्रकार के रेशम के कपड़े, हैंडलूम और प्रिंटेड, क्रमशः 4 : 5 के अनुपात में खरीदे। पार्थ ने दो प्रकार के लिनन कपड़े, A-टाइप और B-टाइप, खरीदे और उसके द्वारा खरीदे गए B-टाइप लिनन की मात्रा, उसके द्वारा खरीदे गए कुल लिनन की $66\frac{2}{3}\%$ है। मिताली द्वारा खरीदे गए हैंडलूम रेशम और पार्थ द्वारा खरीदे गए A-टाइप लिनन की मात्रा के बीच अंतर ज्ञात कीजिए।

- (A) 18
- (B) 22
- (C) 26
- (D) 30
- (E) 24

Q3: Find the ratio of the total quantity of denim fabric purchased by Parth and Mitali together to the total quantity of cotton and linen fabrics purchased by Mitali.

पार्थ और मिताली द्वारा मिलकर खरीदे गए डेनिम कपड़े की कुल मात्रा का, मिताली द्वारा खरीदे गए कपास और लिनन कपड़ों की कुल मात्रा से अनुपात ज्ञात कीजिए।

- (A) 5 : 4
- (B) 8 : 7
- (C) 6 : 5
- (D) 9 : 7
- (E) 7 : 6

Q4: If Mitali purchased denim fabric at Rs. 45 per metre and Parth purchased silk fabric at Rs. 54 per metre, then the total amount spent by Parth on silk fabric is what percent of the total amount spent by Mitali on denim fabric?

यदि मिताली ने डेनिम कपड़ा Rs. 45 प्रति मीटर की दर से खरीदा और पार्थ ने रेशम कपड़ा Rs. 54 प्रति मीटर की दर से खरीदा, तो पार्थ द्वारा रेशम कपड़े पर खर्च की गई कुल राशि, मिताली द्वारा डेनिम कपड़े पर खर्च की गई कुल राशि का कितना प्रतिशत है?

- (A) 75%
- (B) $77 \frac{1}{2}\%$
- (C) $78 \frac{3}{4}\%$
- (D) 80%
- (E) $82 \frac{1}{2}\%$

Q5: The average quantity of cotton, linen and wool fabrics purchased by Parth is 66 metres. The quantity of wool fabric purchased by Mitali is 37.5% more than the quantity of wool fabric purchased by Parth. Find the total quantity of wool fabric purchased by Parth and Mitali together.

पार्थ द्वारा खरीदे गए कपास, लिनन और ऊन के कपड़ों की औसत मात्रा 66 मीटर है। मिताली द्वारा खरीदे गए ऊन के कपड़े की मात्रा, पार्थ द्वारा खरीदे गए ऊन के कपड़े की मात्रा से 37.5% अधिक है। पार्थ और मिताली द्वारा मिलकर खरीदे गए ऊन के कपड़े की कुल मात्रा ज्ञात कीजिए।

- (A) 228
- (B) 216
- (C) 222
- (D) 240
- (E) 234

Solutions:

Fabric	Hindi	Total Quantity	Parth Quantity	Mitali Quantity	Total Check
Cotton	कपास	126	48	78	$48 + 78 = 126$
Silk	रेशम	153	63	90	$63 + 90 = 153$
Linen	लिनन	120	54	66	$54 + 66 = 120$
Denim	डेनिम	168	72	96	$72 + 96 = 168$

1. Option D

Silk purchased by Mitali = $153 - 63 = 90$

Cotton purchased by Kabir = $90 \times 133 \frac{1}{3}\% = 120$

Cotton purchased by Parth = 48

Denim purchased by Parth = 72

Linen purchased by Kabir = $(48 + 72) \div 2 = 60$

Required total = $120 + 60 = 180$

2. Option B

Silk purchased by Mitali = $153 - 63 = 90$

Handloom silk purchased by Mitali = $90 \times 4 \div 9 = 40$

Linen purchased by Parth = 54

B-type linen purchased by Parth = $54 \times 66 \frac{2}{3}\% = 36$

A-type linen purchased by Parth = $54 - 36 = 18$

Required difference = $40 - 18 = 22$

3. Option E

Total denim purchased by Parth and Mitali = 168

Cotton purchased by Mitali = $126 - 48 = 78$

Linen purchased by Mitali = $120 - 54 = 66$

Total cotton and linen purchased by Mitali = $78 + 66 = 144$

Required ratio = $168 : 144 = 7 : 6$

4. Option C

Silk purchased by Parth = 63

Amount spent by Parth on silk = $63 \times 54 = 3402$

Denim purchased by Mitali = $168 - 72 = 96$

Amount spent by Mitali on denim = $96 \times 45 = 4320$

Required percentage = $3402 \times 100 \div 4320 = 78 \frac{3}{4}\%$

5. Option A

Average quantity of cotton, linen and wool purchased by Parth = 66

Total quantity of these three fabrics = $66 \times 3 = 198$

Cotton purchased by Parth = 48

Linen purchased by Parth = 54

Wool purchased by Parth = $198 - 48 - 54 = 96$

Wool purchased by Mitali = $96 \times 137.5\% = 132$

Required total = $96 + 132 = 228$

Set 2: The table shows the total number of coupons available, including redeemed and unused coupons, at five different coupon counters of a trade expo. It also shows the percentage of unused coupons at each counter. Read the table carefully and answer the questions.

तालिका एक व्यापार प्रदर्शनी के पाँच अलग-अलग कूपन काउंटर्स पर उपलब्ध कुल कूपनों की संख्या दर्शाती है, जिसमें रिडीम किए गए और अप्रयुक्त कूपन शामिल हैं। यह प्रत्येक काउंटर पर अप्रयुक्त कूपनों का प्रतिशत भी दर्शाती है। तालिका को ध्यानपूर्वक पढ़िए और प्रश्नों के उत्तर दीजिए।

Note: Total coupons available = Redeemed coupons + Unused coupons.

नोट: उपलब्ध कुल कूपन = रिडीम किए गए कूपन + अप्रयुक्त कूपन।

Coupon Counter कूपन काउंटर	Total coupons available उपलब्ध कुल कूपन	Percentage of unused coupons अप्रयुक्त कूपनों का प्रतिशत
Aarav Counter आरव काउंटर	1350	20%
Bhavya Counter भाव्या काउंटर	882	28 4/7%
Chirag Counter चिराग काउंटर	864	37.5%
Divya Counter दिव्या काउंटर	945	40%
Harit Counter हरित काउंटर	1296	25%

Q1: Find how many more redeemed coupons are there at Chirag Counter than unused coupons at Bhavya Counter.

चिराग काउंटर पर रिडीम किए गए कूपन, भाव्या काउंटर के अप्रयुक्त कूपनों से कितने अधिक हैं?

- (A) 252
- (B) 270
- (C) 306
- (D) 288
- (E) 324

Q2: Total unused coupons at Ishaan Counter are 80% of the total unused coupons at Aarav Counter, and the ratio of total coupons available at Bhavya Counter to that at Ishaan Counter is 7:5. Find the total number of redeemed coupons at Ishaan Counter.

ईशान काउंटर पर कुल अप्रयुक्त कूपन, आरव काउंटर के कुल अप्रयुक्त कूपनों के 80% हैं, और भाव्या काउंटर तथा ईशान काउंटर पर उपलब्ध कुल कूपनों का अनुपात 7:5 है। ईशान

काउंटर पर रिडीम किए गए कुल कूपन ज्ञात कीजिए।

- (A) 396
- (B) 414
- (C) 432
- (D) 450
- (E) 378

Q3: If the ratio of unused coupons at Chirag Counter to the average number of coupons available at Aarav Counter and Bhavya Counter together is $P:Q$, then find $P + Q$.

यदि चिराग काउंटर के अप्रयुक्त कूपनों और आरव काउंटर तथा भाव्या काउंटर पर उपलब्ध कूपनों की औसत संख्या का अनुपात $P:Q$ है, तो $P + Q$ ज्ञात कीजिए।

- (A) 38
- (B) 42
- (C) 44
- (D) 36
- (E) 40

Q4: Unused coupons at Aarav Counter are approximately what percentage of the redeemed coupons at Bhavya Counter?

आरव काउंटर के अप्रयुक्त कूपन, भाव्या काउंटर के रिडीम किए गए कूपनों का लगभग कितने प्रतिशत हैं?

- (A) 43%
- (B) 39%
- (C) 47%
- (D) 36%
- (E) 50%

Q5: Aarav Counter redeemed 55% of its total redeemed coupons for male visitors and the remaining for female visitors. The number of redeemed coupons for female visitors at Harit Counter is 50% more than that at Aarav Counter. Find the total number of redeemed coupons for male visitors at Harit Counter.

आरव काउंटर ने अपने कुल रिडीम किए गए कूपनों में से 55% पुरुष आगंतुकों के लिए और शेष महिला आगंतुकों के लिए रिडीम किए। हरित काउंटर पर महिला आगंतुकों के लिए रिडीम किए गए कूपन, आरव काउंटर की तुलना में 50% अधिक हैं। हरित काउंटर पर पुरुष आगंतुकों के लिए रिडीम किए गए कुल कूपन ज्ञात कीजिए।

- (A) 216

- (B) 252
(C) 243
(D) 270
(E) 225

Solutions:

Coupon Counter कूपन काउंटर	Unused Coupons अप्रयुक्त कूपन	Redeemed Coupons रिडीम किए गए कूपन
Aarav Counter आरव काउंटर	270	1080
Bhavya Counter भाव्या काउंटर	252	630
Chirag Counter चिराग काउंटर	324	540
Divya Counter दिव्या काउंटर	378	567
Harit Counter हरित काउंटर	324	972

Common values:

Aarav Counter: Unused = 20% of 1350 = 270, Redeemed = 1350 - 270 = 1080

Bhavya Counter: Unused = $28 \frac{4}{7}\%$ of 882 = $\frac{2}{7} \times 882 = 252$, Redeemed = 882 - 252 = 630

Chirag Counter: Unused = 37.5% of 864 = $\frac{3}{8} \times 864 = 324$, Redeemed = 864 - 324 = 540

Divya Counter: Unused = 40% of 945 = 378, Redeemed = 945 - 378 = 567

Harit Counter: Unused = 25% of 1296 = 324, Redeemed = 1296 - 324 = 972

1. Option D

Redeemed coupons at Chirag Counter = 540

Unused coupons at Bhavya Counter = 252

Required difference = 540 - 252 = 288

2. Option B

Unused coupons at Ishaan Counter = 80% of 270 = 216

Total coupons at Ishaan Counter = $\frac{5}{7} \times 882 = 630$

Redeemed coupons at Ishaan Counter = 630 - 216 = 414

3. Option E

Unused coupons at Chirag Counter = 324

Average coupons available at Aarav and Bhavya = $(1350 + 882)/2 = 1116$

Ratio = $324:1116 = 9:31$

$P + Q = 9 + 31 = 40$

4. Option A

Unused coupons at Aarav Counter = 270

Redeemed coupons at Bhavya Counter = 630

Required percentage = $270/630 \times 100 = 42.86\% \approx 43\%$

5. Option C

Redeemed coupons at Aarav Counter = 1080

Coupons redeemed for female visitors at Aarav Counter = 45% of 1080 = 486

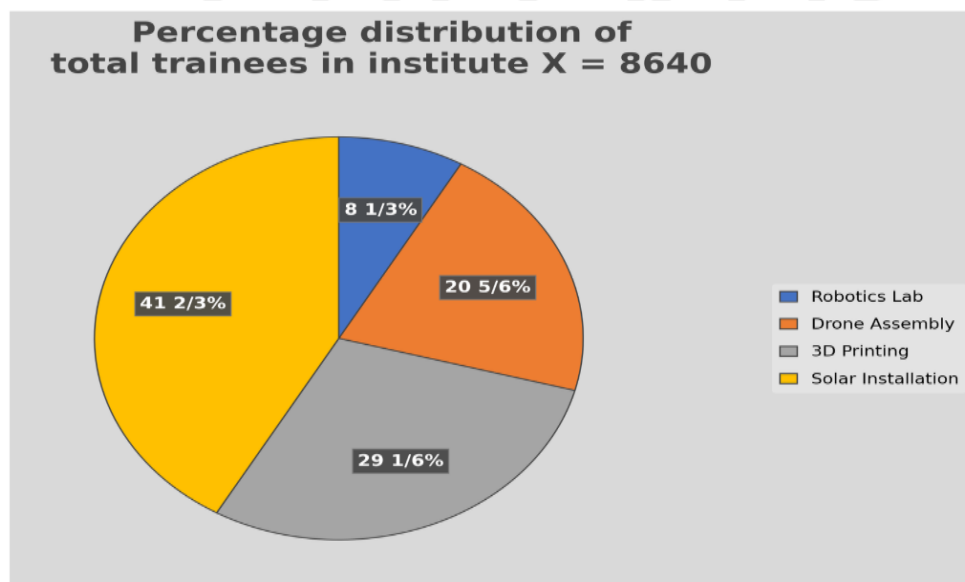
Coupons redeemed for female visitors at Harit Counter = 150% of 486 = 729

Total redeemed coupons at Harit Counter = 972

Coupons redeemed for male visitors at Harit Counter = $972 - 729 = 243$

Set 3: The pie chart shows the percentage distribution of the total number of trainees in a technical skill institute X across four programmes: Robotics Lab, Drone Assembly, 3D Printing and Solar Installation. The total number of trainees in institute X is 8640. The table shows the number of male trainees in each programme. Read the data carefully and answer the following questions.

पाई चार्ट तकनीकी कौशल संस्थान x में चार कार्यक्रमों, रोबोटिक्स लैब, ड्रोन असेंबली, 3D प्रिंटिंग और सोलर इंस्टॉलेशन, में कुल प्रशिक्षुओं का प्रतिशत वितरण दर्शाता है। संस्थान x में कुल प्रशिक्षुओं की संख्या 8640 है। तालिका प्रत्येक कार्यक्रम में पुरुष प्रशिक्षुओं की संख्या दर्शाती है। आँकड़ों को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Programme	Male Trainees
Robotics Lab	450
Drone Assembly	1080
3D Printing	1560
Solar Installation	2400

Q1: If 40% of the female trainees in Robotics Lab are advanced-certificate trainees and the ratio of male trainees to advanced-certificate female trainees in Robotics Lab is 25:6, then find the percentage of non-advanced-certificate trainees out of the total trainees in Robotics Lab.

यदि रोबोटिक्स लैब में 40% महिला प्रशिक्षु एडवांस्ड-सर्टिफिकेट प्रशिक्षु हैं और रोबोटिक्स लैब में पुरुष प्रशिक्षुओं तथा एडवांस्ड-सर्टिफिकेट महिला प्रशिक्षुओं का अनुपात 25:6 है, तो रोबोटिक्स लैब के कुल प्रशिक्षुओं में से गैर-एडवांस्ड-सर्टिफिकेट प्रशिक्षुओं का प्रतिशत ज्ञात कीजिए।

- (A) 78%
- (B) 82%
- (C) 90%
- (D) 85%
- (E) 75%

Q2: If the ratio of male trainees to female trainees in institute Y is 5:4 and the number of male trainees in institute Y is 150% more than the number of male trainees in Drone Assembly of institute X, then find the ratio of total trainees in institute Y to total trainees in institute X.

यदि संस्थान Y में पुरुष प्रशिक्षुओं और महिला प्रशिक्षुओं का अनुपात 5:4 है और संस्थान Y में पुरुष प्रशिक्षुओं की संख्या, संस्थान X के ड्रोन असेंबली कार्यक्रम के पुरुष प्रशिक्षुओं की संख्या से 150% अधिक है, तो संस्थान Y के कुल प्रशिक्षुओं का संस्थान X के कुल प्रशिक्षुओं से अनुपात ज्ञात कीजिए।

- (A) 7:16
- (B) 11:18
- (C) 9:16
- (D) 5:12
- (E) 13:24

Q3: If 20% of the female trainees from Solar Installation leave Solar Installation and join 3D Printing, then find the percentage increase in the number of female trainees in 3D Printing.

यदि सोलर इंस्टॉलेशन की 20% महिला प्रशिक्षु सोलर इंस्टॉलेशन छोड़कर 3D प्रिंटिंग में शामिल हो जाती हैं, तो 3D प्रिंटिंग में महिला प्रशिक्षुओं की संख्या में प्रतिशत वृद्धि ज्ञात कीजिए।

- (A) 25%
- (B) 30%
- (C) 20%
- (D) 35%
- (E) 40%

Q4: If the number of male trainees in institute K is 480 more than the number of female trainees in Drone Assembly of institute X, then find how much less the average number of female trainees in Robotics Lab, 3D Printing and Solar Installation is than the number of male trainees in institute K.

यदि संस्थान K में पुरुष प्रशिक्षुओं की संख्या, संस्थान X के ड्रोन असेंबली कार्यक्रम की महिला

प्रशिक्षुओं की संख्या से 480 अधिक है, तो रोबोटिक्स लैब, 3D प्रिंटिंग और सोलर इंस्टॉलेशन में महिला प्रशिक्षुओं की औसत संख्या, संस्थान K के पुरुष प्रशिक्षुओं की संख्या से कितनी कम है?

- (A) 330
- (B) 360
- (C) 420
- (D) 450
- (E) 390

Q5: If the ratio of trainees preparing for Government technical jobs to Private technical jobs in Robotics Lab, Drone Assembly, 3D Printing and Solar Installation is 5:7, 7:5, 5:4 and 4:5 respectively, then find the ratio of total trainees preparing for Government technical jobs to those preparing for Private technical jobs in institute X.

यदि रोबोटिक्स लैब, ड्रोन असेंबली, 3D प्रिंटिंग और सोलर इंस्टॉलेशन में सरकारी तकनीकी नौकरियों की तैयारी करने वाले प्रशिक्षुओं का निजी तकनीकी नौकरियों की तैयारी करने वाले प्रशिक्षुओं से अनुपात क्रमशः 5:7, 7:5, 5:4 और 4:5 है, तो संस्थान X में सरकारी तकनीकी नौकरियों की तैयारी करने वाले कुल प्रशिक्षुओं का निजी तकनीकी नौकरियों की तैयारी करने वाले कुल प्रशिक्षुओं से अनुपात ज्ञात कीजिए।

- (A) 143:145
- (B) 145:143
- (C) 147:145
- (D) 149:143
- (E) 151:147

Solutions:

Programme	Percentage	Total trainees	Male trainees	Female trainees
Robotics Lab	8 1/3%	720	450	270
Drone Assembly	20 5/6%	1800	1080	720
3D Printing	29 1/6%	2520	1560	960
Solar Installation	41 2/3%	3600	2400	1200
Total	100%	8640	5490	3150

Common Calculation:

Total trainees in institute X = 8640

Robotics Lab trainees = $8\frac{1}{3}\%$ of 8640 = 720

Drone Assembly trainees = $20\frac{5}{6}\%$ of 8640 = 1800

3D Printing trainees = $29\frac{1}{6}\%$ of 8640 = 2520

Solar Installation trainees = $41\frac{2}{3}\%$ of 8640 = 3600

Female trainees in Robotics Lab = 720 – 450 = 270

Female trainees in Drone Assembly = 1800 – 1080 = 720

Female trainees in 3D Printing = 2520 – 1560 = 960

Female trainees in Solar Installation = 3600 – 2400 = 1200

1. Option D

Female trainees in Robotics Lab = 270

Advanced-certificate female trainees = 40% of 270 = 108

Non-advanced-certificate trainees = 270 – 108 = 162

Required percentage = $\frac{162}{270} \times 100 = 60\%$

2. Option C

Male trainees in Drone Assembly = 1080

Male trainees in institute Y = $1080 \times \frac{250}{100} = 2700$

Male:female in institute Y = 5:4

Total trainees in institute Y = $2700 \times \frac{9}{5} = 4860$

Required ratio = 4860:8640 = 9:16

3. Option A

Female trainees in Solar Installation = 1200

Female trainees shifted to 3D Printing = 20% of 1200 = 240

Female trainees in 3D Printing = 960

Percentage increase = $\frac{240}{960} \times 100 = 25\%$

4. Option E

Female trainees in Drone Assembly = 720

Male trainees in institute K = 720 + 480 = 1200

Average female trainees in Robotics Lab, 3D Printing and Solar Installation = $\frac{(270 + 960 + 1200)}{3} = 810$

Required difference = $1200 - 810 = 390$

5. Option B

Government and Private in Robotics Lab = 5:7, so Government = 300 and Private = 420

Government and Private in Drone Assembly = 7:5, so Government = 1050 and Private = 750

Government and Private in 3D Printing = 5:4, so Government = 1400 and Private = 1120

Government and Private in Solar Installation = 4:5, so Government = 1600 and Private = 2000

Total Government = $300 + 1050 + 1400 + 1600 = 4350$

Total Private = $420 + 750 + 1120 + 2000 = 4290$

Required ratio = $4350:4290 = 145:143$

Set 4: The table shows the number of seed packets packed by three nurseries, Narmada Nursery, Kaveri Nursery and Godavari Nursery, in four different seasonal batches. Some data is missing. Wherever a percentage is given, it represents that nursery's percentage share out of the total seed packets packed in that batch. Read the data carefully and answer the following questions.

तालिका चार अलग-अलग मौसमी बैचों में तीन नर्सरियों, नर्मदा नर्सरी, कावेरी नर्सरी और गोदावरी नर्सरी द्वारा पैक किए गए बीज पैकेटों की संख्या को दर्शाती है। कुछ डेटा लुप्त है। जहाँ प्रतिशत दिया गया है, वह उस बैच में पैक किए गए कुल बीज पैकेटों में उस नर्सरी का प्रतिशत हिस्सा दर्शाता है। डेटा को ध्यानपूर्वक पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।

Period (अवधि)	Narmada Nursery (नर्मदा नर्सरी)	Kaveri Nursery (कावेरी नर्सरी)	Godavari Nursery (गोदावरी नर्सरी)	Total Seed Packets (कुल बीज पैकेट)
Spring Batch (वसंत बैच)	1100	--	40%	5500
Monsoon Batch (मानसून बैच)	1320	25%	--	4400
Autumn Batch (शरद बैच)	--	33.33%	1760	6600
Winter Batch (शीतकालीन बैच)	2860	1540	1980	--

Q1: The simplified ratio between the number of seed packets packed by Kaveri Nursery in Autumn Batch and the number of seed packets packed by Narmada Nursery in Winter Batch is $p : q$. Find the value of $(p + q)$.

शरद बैच में कावेरी नर्सरी द्वारा पैक किए गए बीज पैकेटों और शीतकालीन बैच में नर्मदा नर्सरी द्वारा पैक किए गए बीज पैकेटों का सरल अनुपात $p : q$ है। $(p + q)$ का मान ज्ञात कीजिए।

- (A) 19
- (B) 23
- (C) 21
- (D) 25
- (E) 27

Q2: Find the total number of seed packets packed by Godavari Nursery in all four batches together.

गोदावरी नर्सरी द्वारा चारों बैचों में मिलाकर पैक किए गए बीज पैकेटों की कुल संख्या ज्ञात कीजिए।

- (A) 7480
- (B) 7700
- (C) 8140

- (D) 8360
- (E) 7920

Q3: The number of seed packets packed by Godavari Nursery in Monsoon Batch is what percent more than the number of seed packets packed by Kaveri Nursery in Monsoon Batch? Find the percentage value.

मानसून बैच में गोदावरी नर्सरी द्वारा पैक किए गए बीज पैकेट, मानसून बैच में कावेरी नर्सरी द्वारा पैक किए गए बीज पैकेटों से कितने प्रतिशत अधिक हैं? प्रतिशत मान ज्ञात कीजिए।

- (A) 80
- (B) 70
- (C) 60
- (D) 90
- (E) 100

Q4: In Autumn Batch, Kaveri Nursery packed 220 damaged seed packets. The ratio of damaged seed packets packed by Kaveri Nursery to Narmada Nursery in Autumn Batch is 5 : 4. Find the number of non-damaged seed packets packed by Narmada Nursery in Autumn Batch.

शरद बैच में कावेरी नर्सरी ने 220 खराब बीज पैकेट पैक किए। शरद बैच में कावेरी नर्सरी और नर्मदा नर्सरी द्वारा पैक किए गए खराब बीज पैकेटों का अनुपात 5 : 4 है। शरद बैच में नर्मदा नर्सरी द्वारा पैक किए गए गैर-खराब बीज पैकेटों की संख्या ज्ञात कीजिए।

- (A) 2398
- (B) 2508
- (C) 2420
- (D) 2464
- (E) 2552

Q5: Find the average number of seed packets packed by all three nurseries in Autumn Batch.

शरद बैच में तीनों नर्सरियों द्वारा पैक किए गए बीज पैकेटों की औसत संख्या ज्ञात कीजिए।

- (A) 1980
- (B) 2310
- (C) 2200
- (D) 2420
- (E) 2090

Solutions:

Period (अवधि)	Narmada (नर्मदा)	Kaveri (कावेरी)	Godavari (गोदावरी)	Total (कुल)
Spring Batch (वसंत बैच)	1100	2200	2200	5500
Monsoon Batch (मानसून बैच)	1320	1100	1980	4400
Autumn Batch (शरद बैच)	2640	2200	1760	6600
Winter Batch (शीतकालीन बैच)	2860	1540	1980	6380

1. Option B

Kaveri Nursery in Autumn Batch = 33.33% of 6600 = $\frac{1}{3} \times 6600 = 2200$

Narmada Nursery in Winter Batch = 2860

Required ratio = 2200 : 2860 = 10 : 13

$p + q = 10 + 13 = 23$

2. Option E

Godavari Nursery in Spring Batch = 40% of 5500 = 2200

Kaveri Nursery in Monsoon Batch = 25% of 4400 = 1100

Godavari Nursery in Monsoon Batch = 4400 - 1320 - 1100 = 1980

Godavari Nursery in Autumn Batch = 1760

Godavari Nursery in Winter Batch = 1980

Total = 2200 + 1980 + 1760 + 1980 = 7920

3. Option A

Kaveri Nursery in Monsoon Batch = 25% of 4400 = 1100

Godavari Nursery in Monsoon Batch = 4400 - 1320 - 1100 = 1980

Required percent more = $\frac{(1980 - 1100)}{1100} \times 100$

Required percent more = $\frac{880}{1100} \times 100 = 80$

4. Option D

Kaveri damaged packets = 220

Damaged Kaveri : Damaged Narmada = 5 : 4

Damaged Narmada packets = $220 \times \frac{4}{5} = 176$

Narmada Nursery in Autumn Batch = 6600 - 2200 - 1760 = 2640

Non-damaged Narmada packets = 2640 - 176 = 2464

5. Option C

Narmada Nursery in Autumn Batch = 2640

Kaveri Nursery in Autumn Batch = 2200

Godavari Nursery in Autumn Batch = 1760

Average = $(2640 + 2200 + 1760) / 3$

Average = $6600 / 3 = 2200$

CHECKLIST
BY
AASHISH
ARORA